

JOHN ROSS M. RIVERA

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SUMMARY

Full-stack software engineer (TypeScript, NestJS, React/Next.js) who owns production systems end-to-end, and author of **provenly**, an open-source verification and guardrail layer that makes AI coding agents prove their work. Primary developer of a competition platform's admin console, backend APIs, and payment stack; sole builder of a Laravel affiliate/payout portal.

WORK EXPERIENCE

Crystal Vision - Software Engineer, Full-Stack | Remote

Aug 2025 - Present

Rola Competition Platform (Full Stack)

- Let operations staff run the full competition lifecycle (judging, moderation, payments, analytics) without engineering involvement, as sole developer of the admin console (**34 modules, 47 pages**; Next.js/React).
- Primary developer of the backend powering both the admin console and the mobile app: **37 modules, 200+ REST endpoints, 61 data models** (NestJS).
- Enabled crypto and fiat entry payments **without double-charging users**, integrating **2 payment rails** (USDC on Polygon + Xendit).
- Protected contest-result integrity against vote manipulation and race conditions under load, with a **concurrency-safe voting engine** (per-user and per-IP rate limiting).
- Gave the business self-serve visibility into users, revenue, and engagement with **9+ analytics dashboards** and CSV export.
- Cut team-wide review and debugging time with an in-house **Claude Code agent harness** (8 subagents, secret-scanning and pre-commit guardrails), since generalised and open-sourced, below.

KOL Dashboard (Full Stack)

- Gave the company one system to run its entire **KOL/affiliate program**, owning a Laravel 12 portal with a schema designed from scratch (**16 models, 23 migrations**).
- Automated affiliate commission and payout approvals so finance stopped reconciling by hand, with a payout state machine and role-based approval across **4 access levels**.
- Kept emails, names, and wallet addresses **out of the UI, logs, and exports** with a reusable masking layer (25 unit tests).

RR Remo Trucking - Software Engineer, Full-Stack | Freelance

May - June 2026

- Built an AI-assisted logistics site (Next.js/React, Groq LLaMA 3.1) that automated customer support and lead capture.
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OPEN SOURCE

provenly - Author & Maintainer | github.com/Heisenberg18/provenly

2026

Reliability infrastructure for AI coding agents. MIT, zero dependencies, any language, installed with one npx command.

- **Solved the "false green" problem**: the failure mode where an agent reports tests passed without running them. An append-only ledger records every file edit and verification command with its exit code, and a Stop-hook claim checker **blocks any turn asserting a green gate the record cannot support**.
 - **Escalates on stuck failures, not retry counts**: fingerprints each failure set so converging fixes run uninterrupted while a thrashing loop halts at 3 with a written report. A plan verifier separately **grades each step's own check** (real / self / exists), catching steps that "pass" by proving authorship rather than behaviour.
 - Contained an agent that runs unattended by **removing the shell** from the path that executes model-authored commands, and made role separation real with **5 PreToolUse guards** (destructive commands, secret leaks, unsafe commits, out-of-scope writes, cross-session file locks) enforcing **5 scoped subagents and 8 skills** across **36 scripts, ~8,300 LOC**.
 - Closed the gap between *configured* and *firing*: a hook heartbeat plus selftest fails on any hook registered but never observed, backed by **751 assertions across 20 suites** testing the ALLOW path as rigorously as the BLOCK path.
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EDUCATION

Bulacan State University - BS Information Technology

2020 - 2025

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, PHP, SQL, HTML, CSS

Frameworks: NestJS, Node.js, Express, Next.js, React, Redux Toolkit, RTK Query, Laravel, Tailwind CSS

AI Engineering: Claude Code (hooks, subagents, skills), agent guardrails & evaluation, LLM tool orchestration, Groq / LLaMA, prompt engineering

Developer Tools: MySQL, Prisma, Sequelize, Redis, JWT, RBAC, Jest, Playwright, Pest / PHPUnit, Git, GitHub, Docker, AWS S3, Postman, Jira, CI/CD